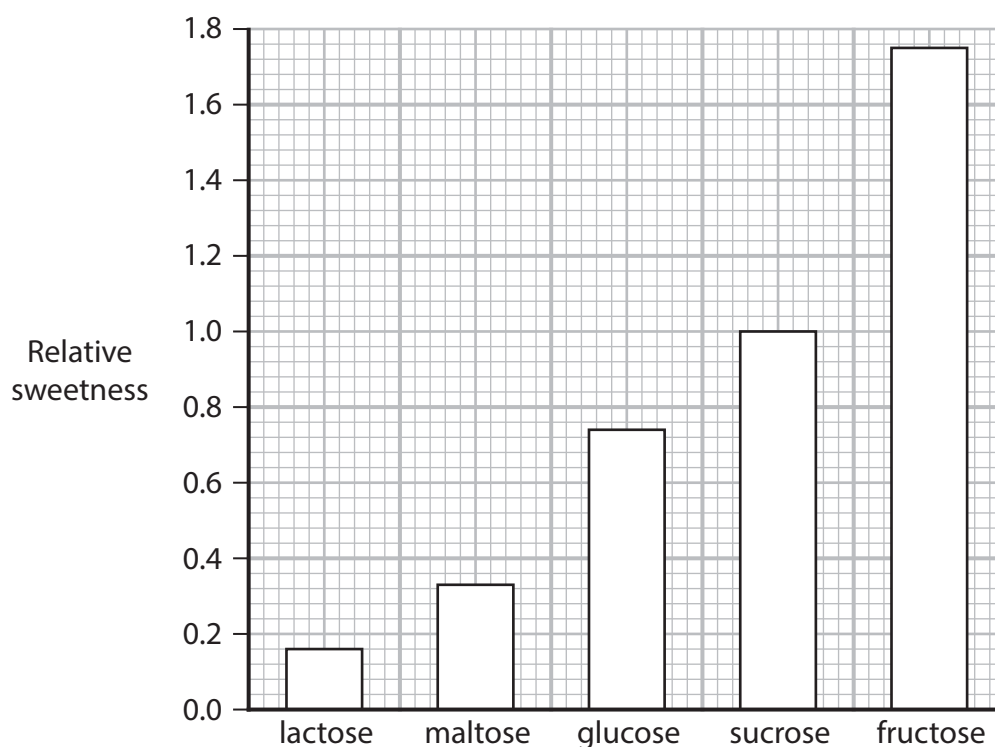


4 Carbohydrates are important molecules in living organisms.

- (a) The graph shows the relative sweetness of some monosaccharides and some disaccharides.



- (i) Which row in the table gives the correct number of monosaccharides and disaccharides shown in this graph?

(1)

	Number of monosaccharides	Number of disaccharides
☐ A	1	4
☐ B	2	3
☐ C	3	2
☐ D	4	1

- (ii) Which bond joins two monosaccharides together to form a disaccharide?

(1)

- ☐ A ester
- ☐ B glycosidic
- ☐ C hydrogen
- ☐ D phosphodiester



(iii) State **two** conclusions that can be made about the relative sweetness of monosaccharides and of disaccharides.

(2)

1

.....

.....

2

.....

.....

(iv) Devise an investigation that a student could carry out to confirm the data shown in the graph.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(b) Amylose and amylopectin are components of starch.

Compare and contrast the structure of amylose with the structure of amylopectin.

(3)

(Total for Question 4 = 10 marks)

